

A practical roadmap to building, upgrading & future-proofing your HR Analytics career

PAGE 1 — HR ANALYTICS IS NOT JUST DASHBOARDS

A Power BI dashboard doesn't automatically make you an HR Analytics professional.

The real process is:

HR Data → Question → Analysis → Insight → Recommendation → Action → Measurement

The goal isn't to make prettier charts.

The goal is to help HR and business leaders make **better-informed decisions using people data**.

PAGE 2 — UNDERSTAND HR ANALYTICS

HR Analytics can cover:

- Workforce Analytics
- People Analytics
- Recruitment Analytics
- Attrition Analytics
- Compensation Analytics
- Performance Analytics
- Diversity Analytics
- Learning Analytics
- Engagement Analytics
- Workforce Planning

Learn from

- CIPD
- SHRM
- LinkedIn Learning
- Coursera
- YouTube

- Microsoft Learn
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PAGE 3 — START WITH HR, NOT DATA

Before becoming an analyst, understand HR.

Learn:

- Recruitment
- HR Operations
- Payroll
- Performance
- Compensation
- L&D
- Employee Relations
- Attrition
- Workforce planning

Why?

Because a technically correct analysis can still be **irrelevant to the business** if you don't understand the HR problem.

PAGE 4 — EXCEL IS YOUR FOUNDATION

Master:

Basic

- Sorting
- Filtering
- Data cleaning
- Formatting

Intermediate

- XLOOKUP/VLOOKUP
- IF/IFS
- COUNTIF/COUNTIFS

- SUMIF/SUMIFS
- Date functions
- Text functions

Advanced

- Pivot Tables
- Power Query
- Data validation
- Error handling
- Dashboards

Learn from

- Microsoft Learn
- ExcelJet
- LinkedIn Learning
- Coursera
- YouTube

PAGE 5 — DATA CLEANING

Real HR data is rarely perfect.

Learn to identify:

- Missing values
- Duplicate records
- Incorrect dates
- Inconsistent departments
- Different naming conventions
- Invalid employee IDs
- Outliers
- Incorrect calculations

Practice

Create a fictional employee dataset of **500 employees**.

Clean it and document:

What was wrong → What you changed → Why it mattered.

PAGE 6 — HR METRICS

Learn what the numbers actually mean.

Recruitment

- Time to hire
- Time to fill
- Source of hire
- Offer acceptance
- Joining ratio

Workforce

- Headcount
- FTE
- Attrition
- Absenteeism
- Tenure

Compensation

- Salary distribution
- Salary growth
- Compensation cost

Performance

- Rating distribution
 - Performance trends
 - Promotion patterns
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PAGE 7 — STATISTICS FUNDAMENTALS

You don't need to become a mathematician.

But understand:

- Mean
- Median
- Mode
- Percentage
- Variance
- Standard deviation
- Distribution
- Correlation
- Sampling
- Outliers
- Basic probability

Later:

- Regression
- Hypothesis testing
- Predictive analytics

Learn from

- Khan Academy
- Coursera
- LinkedIn Learning
- YouTube

PAGE 8 — SQL

SQL becomes increasingly valuable as datasets become larger.

Learn:

- SELECT
- WHERE
- GROUP BY
- ORDER BY
- JOIN
- CASE WHEN
- COUNT
- SUM
- AVG
- Subqueries

- CTEs
- Window functions

Learn from

- SQLBolt
- W3Schools
- Coursera
- DataCamp
- YouTube

Practice

Create employee tables for:

Employees + Departments + Salary + Performance + Attrition

Then answer business questions using SQL.

PAGE 9 — POWER BI

Power BI is one of the most useful tools for HR/People Analytics.

Learn:

- Data import
- Data cleaning
- Relationships
- Data modelling
- DAX basics
- Measures
- Filters
- Visualisations
- Drill-down
- Dashboard design

Learn from

Microsoft Learn should be your primary starting point.

Also:

- Coursera
 - LinkedIn Learning
 - YouTube
-

PAGE 10 — BUILD HR DASHBOARDS

Don't build dashboards just to demonstrate charts.

Build around business questions.

Dashboard 1 — Workforce

Headcount → Department → Location → Gender → Tenure

Dashboard 2 — Attrition

Attrition → Department → Tenure → Role → Manager → Location

Dashboard 3 — Recruitment

Applications → Screening → Interview → Offer → Joining

Dashboard 4 — Compensation

Salary → Department → Level → Gender → Location

PAGE 11 — DATA STORYTELLING

This is where many analysts struggle.

Don't say:

"Attrition is 18%."

Say:

"Attrition increased from X to Y, with the largest change occurring in this population. The data suggests these areas require further investigation."

Learn to communicate:

What happened?

Why might it have happened?

Who is affected?

What should the business investigate?

What action can be considered?

PAGE 12 — CORRELATION $\neq$ CAUSATION

This is critical in People Analytics.

If employees with shorter tenure leave more frequently, you cannot automatically conclude:

"Short tenure causes attrition."

You need to investigate other factors.

Learn:

- Correlation
- Causation
- Confounding variables
- Bias
- Data limitations

Learn from

- Khan Academy
- Coursera
- DataCamp
- LinkedIn Learning

PAGE 13 — RECRUITMENT ANALYTICS

Learn to analyse the hiring funnel:

Application → Screening → Interview → Offer → Acceptance → Joining

Track:

- Source effectiveness
- Time to hire
- Recruiter productivity
- Offer acceptance
- Joining ratio
- Cost per hire
- Candidate drop-off

Portfolio project

Create a recruitment dashboard and provide **5 actionable insights**.

PAGE 14 — ATTRITION ANALYTICS

Analyse:

- Overall attrition
- Voluntary vs involuntary
- Department
- Location
- Tenure
- Level
- Manager
- Compensation
- Performance
- Internal movement

But remember:

Data can identify patterns. It doesn't automatically explain employee motives.

Use surveys, interviews and other evidence where appropriate.

PAGE 15 — COMPENSATION ANALYTICS

Understand:

- Salary distribution
- Internal equity
- Salary ranges
- Compa-ratio
- Salary increases
- Variable pay
- Compensation cost
- Pay progression

Learn the basics of:

Compensation + Finance + Statistics + HR.

Learn from

- WorldatWork resources
 - LinkedIn Learning
 - Coursera
 - YouTube
-

PAGE 16 — WORKFORCE PLANNING

Move beyond reporting current headcount.

Understand:

Current workforce → Future requirement → Gap → Hiring/Development/Movement plan

Analyse:

- Headcount trends
- Workforce cost
- Attrition

- Hiring demand
- Skills
- Critical roles
- Capacity

Learn from

- CIPD
 - SHRM
 - Coursera
 - LinkedIn Learning
 - YouTube
-

PAGE 17 — PEOPLE ANALYTICS & AI

AI can help with:

- Data exploration
- Formula generation
- SQL assistance
- DAX assistance
- Data summarisation
- Pattern identification
- Report writing
- Dashboard narratives
- Scenario analysis

Tools:

- ChatGPT
- Microsoft Copilot
- Gemini
- Power BI Copilot/features where available

Golden rule

AI can assist analysis. You remain responsible for validating the result.

Never upload identifiable employee data, salary information, health information or confidential company data into public AI tools.

PAGE 18 — DATA PRIVACY & ETHICS

HR data is sensitive.

Understand:

- Data minimisation
- Access controls
- Anonymisation
- Confidentiality
- Data governance
- Appropriate use
- Bias
- Fairness
- Employee privacy

Learn from

- Your organisation's data policies
- Official privacy/data-protection resources applicable to your jurisdiction
- CIPD
- SHRM
- Microsoft security/privacy resources

PAGE 19 — 0–2 YEARS

Focus on:

- HR fundamentals
- Excel
- Data cleaning
- HR metrics
- Basic statistics
- Reporting
- Power BI basics
- Data storytelling

Target roles

- HR MIS Analyst

- HR Reporting Analyst
 - HR Analyst
 - People Analytics Analyst
 - Workforce Reporting Analyst
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PAGE 20 — 2–5 YEARS

Move towards:

- SQL
- Advanced Power BI
- Data modelling
- Workforce analytics
- Attrition analytics
- Compensation analytics
- Recruitment analytics
- Stakeholder consulting
- Predictive analytics

Possible roles

- People Analytics Specialist
 - HR Analytics Consultant
 - Workforce Analytics Specialist
 - People Insights Analyst
 - Senior HR Analyst
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PAGE 21 — 5+ YEARS

Potential direction:

Senior Analyst → People Analytics Manager → People Analytics Lead → Head of People Analytics/Insights

Build:

- Analytics strategy
- Workforce planning
- Predictive modelling

- Executive reporting
 - Data governance
 - People insights
 - HR technology
 - Strategic workforce decisions
 - Team leadership
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PAGE 22 — SPECIALISATIONS

You can specialise in:

People Analytics

Broad people-data insights.

Workforce Analytics

Headcount, capacity, workforce planning.

Talent Analytics

Performance, talent, succession and mobility.

Compensation Analytics

Pay, rewards and workforce cost.

Recruitment Analytics

Hiring funnel and talent acquisition data.

Learning Analytics

Training, skills and capability data.

People Consulting

Use analytics to solve broader organisational problems.

PAGE 23 — BUILD YOUR PORTFOLIO

Create at least 5 projects:

1. Workforce Dashboard

500 employees.

2. Attrition Analysis

Identify patterns and limitations.

3. Recruitment Funnel

Analyse hiring conversion.

4. Compensation Dashboard

Salary distribution and trends.

5. Workforce Planning Model

Forecast workforce requirements using a fictional dataset.

For each project show:

Problem → Data → Method → Insight → Recommendation → Limitation

PAGE 24 — CV FOR HR ANALYTICS

Don't write:

"Created HR dashboards."

Write:

Built an interactive Power BI workforce dashboard covering 2,500 employees across 6 locations, enabling department-wise headcount, attrition and tenure analysis.

Show:

Scale + Tool + Analysis + Business question + Impact

PAGE 25 — 90-DAY ANALYTICS PLAN

Days 1–30

Master:

- HR metrics
- Excel
- Data cleaning
- Basic statistics

Days 31–60

Learn:

- Power BI
- Data modelling
- SQL basics
- Data storytelling

Build:

Workforce + Attrition dashboard

Days 61–90

Build:

- Recruitment dashboard
- Compensation analysis
- Workforce planning project

Then update:

CV + LinkedIn + Portfolio

PAGE 26 — HR ANALYTICS SELF-CHECK

Can you:

- ☐ Clean HR data?
- ☐ Use Excel confidently?
- ☐ Calculate HR metrics?
- ☐ Understand basic statistics?
- ☐ Write SQL queries?
- ☐ Build Power BI dashboards?
- ☐ Explain a business insight?
- ☐ Understand correlation vs causation?
- ☐ Handle sensitive data responsibly?
- ☐ Build a portfolio?
- ☐ Speak to HR/business stakeholders?

PAGE 27 — BIGGEST ANALYTICS CAREER MISTAKE

Don't become:

"The person who makes the HR dashboard."

Become:

The person who can understand the HR problem, analyse the data, explain the insight and help stakeholders decide what to investigate or do next.

The evolution is:

Reporting → Analytics → Insights → Strategic Decision Support

PAGE 28 — FINAL WORD

HR Analytics sits at the intersection of:

HR + Data + Technology + Statistics + Business

You don't need to master everything on Day 1.

Start with:

**HR fundamentals → Excel → Statistics → Power BI → SQL → Storytelling → Business Acumen
→ Advanced Analytics/AI**

That progression gives you a clear path from **HR reporting to People Analytics.**